

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Magesh. S
Register Number	192525002

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I MAGESH. S (192525002) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

EXP NO 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM: To Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

Step 4: Enter application name as library book reservation system.

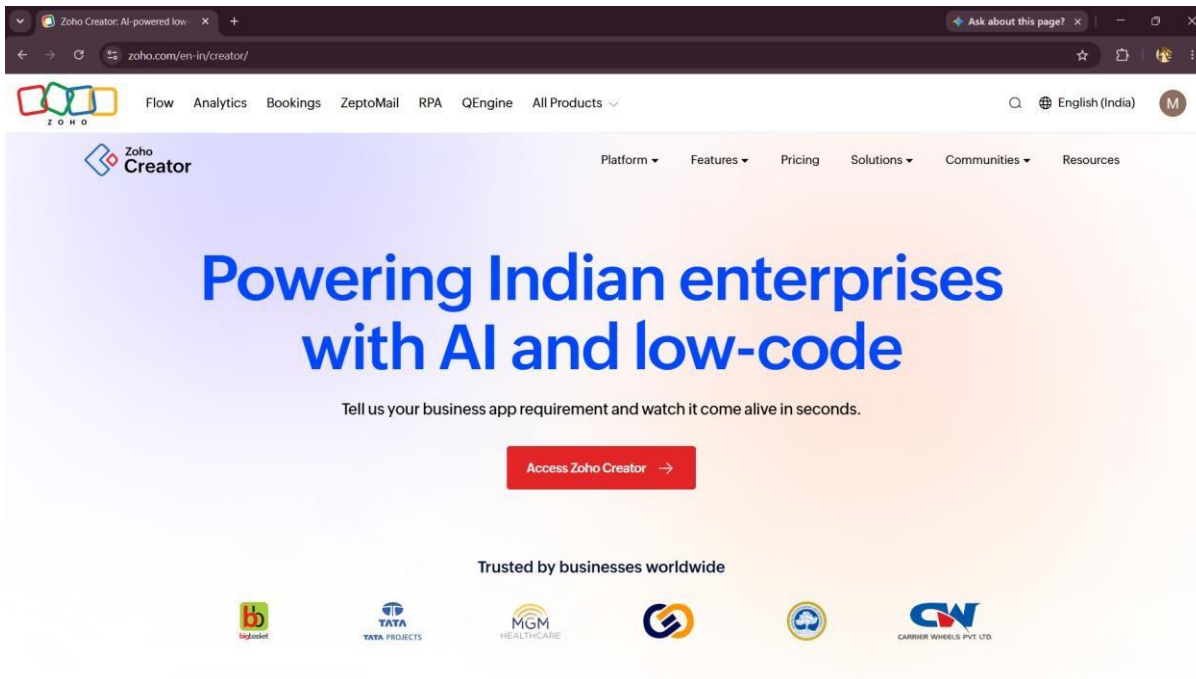
step 5: Created new application as library book reservation system.

step 6: Select one form

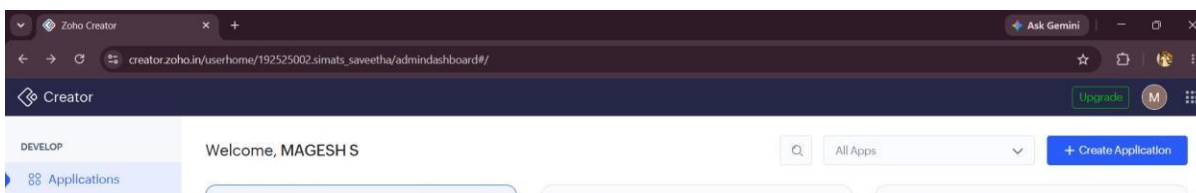
step 7: The software has been created.

IMPLEMENTATION:

STEP1: GOTO [ZHO.COM](https://zoho.com)



STEP 2: LOGINTOTHE ZOHO.COM



STEP 3: SELECT ONE APPLICATION

Create New Application



Create using Zia
Build a functional application effortlessly using GenAI

Create



Create from scratch
Create a new application for your business needs

Create



Create from gallery
Pick a pre-built application that suits your requirement

Pick



Import from file
Import your existing data to create an application

Import

STEP 4: ENTER APPLICATION NAME

×



Create from scratch

Application Name *

Enter your application name

☐ Enable Environment [Learn More](#)

Create Application

STEP 5: CREATED NEW APPLICATION

PROPERTY BOOKING AND RENTAL

PROPERTY DETAILS

Done

Basic Fields

Name

Email

Address

Single Line

Multi Line

123

Number

Date

Time

Drop Down

Phone

PROPERTY ID

OWNER NAME

Phone

Email

PROPERTY TYPE

PURPOSE

ADDRESS

AREA

CITY

LOCALITY

Field Properties

Field name

DESCRIPTION

Field link name

Multi_Select1

View Field References

Choices

Advanced

Renaming the choices will update the values in the existing records, if any.

Choice 1

Choice 2

Choice 3

Alphabetical Order

Validation

STEP 6: SELECT ONE FORM

How would you like to create your form?

On my own

From scratch

Import Schema

Using Zia

From a template

Cab Booking

Conf Room Booking

Appointment Booking

Hotel Room Booking

Complaints

50 + View More

STEP 7: THE SOFTWARE HAS BEEN CREATED.

The screenshot displays the SIMATS LIBRARY application interface. The sidebar on the left contains navigation options: SIMATS ENGINEERING, Authors, SIMATS ENGINEERING Report (highlighted), Publishers, Books, Copies, Members, Reservations, and Loans. The main content area is titled 'SIMATS ENGINEERING Report' and features a table with the following data:

Author ID	Author Name	Country	Notes
4	Thalapathy	India	
3	Vijay	India	
2	Vikam David	India	
1	Magesh		

The bottom status bar indicates 'Showing 4 of 4'.

RESULT:

Thus, the **SIMATS Library Book Reservation System** was successfully created using **Zoho Creator** as a cloud service provider, demonstrating the **SaaS model**. The application allows library book details to be stored, managed, and accessed through the cloud.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM:

To create a simple cloud software application for Payroll Processing System using Zoho Creator as a Cloud Service Provider to demonstrate the Software as a Service (SaaS) model.

PROCEDURE:

Step 1: Go to zoho.com.

Step 2: Log in to your Zoho account.

Step 3: Open Zoho Creator from the Zoho applications.

Step 4: Click Create New Application.

Step 5: Enter the application name as Payroll Processing System.

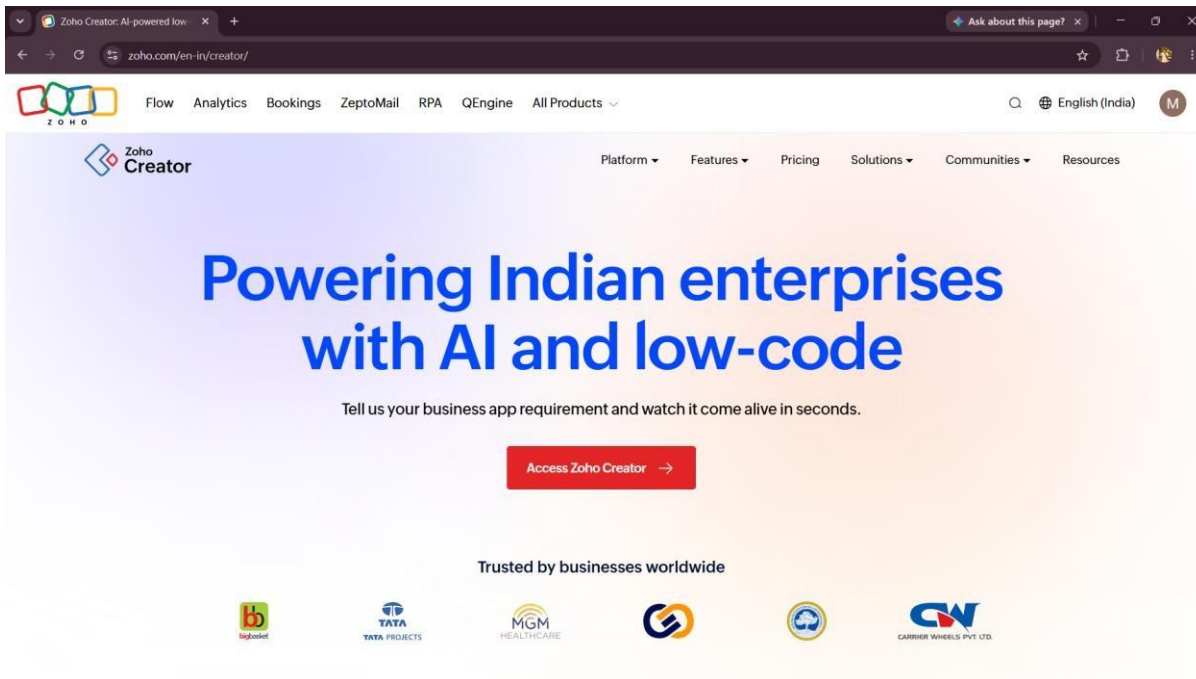
Step 6: Create a new application.

Step 7: Create a new form named Employee Payroll Details.

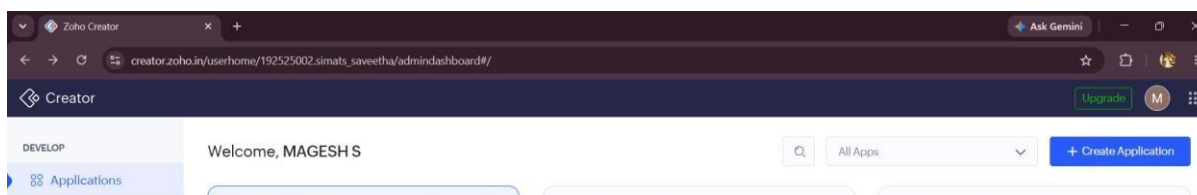
Step 8: The Payroll Processing System software has been successfully created and deployed on the Zoho cloud platform.

IMPLEMENTATION:

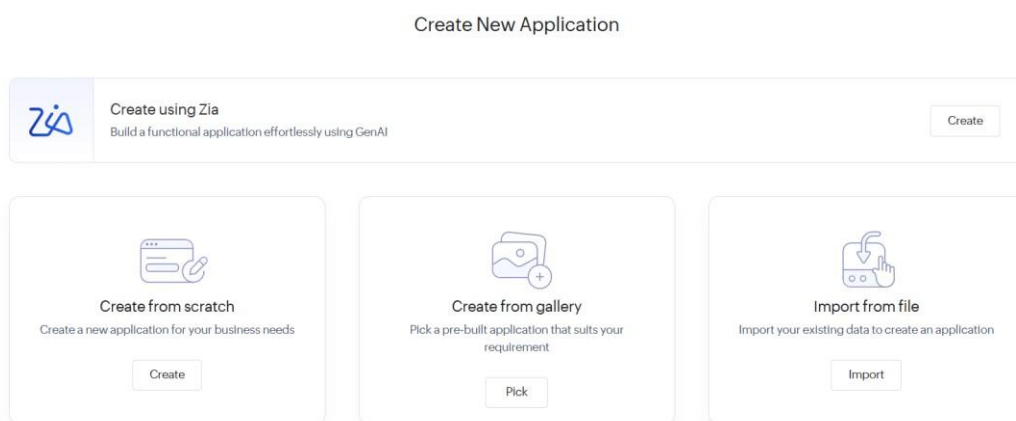
STEP1: GOTO Zoho.COM



STEP 2: LOGINTOTHE ZOHO.COM



STEP 3: SELECT ONE APPLICATION



×

STEP 5: CREATED NEW APPLICATION

192525002.simat.s.aveetha -

192525002.simat.s.aveetha/payroll-processing-system1#Report:payroll_records_Report

Click to go forward, hold to see history

Trial has expired Upgrade

Edit this application

Help

Payroll Processing System

Dashboard

Employees

Payroll Records

Payroll Records Re...

Board

Pay Components

Allowance Config...

Payroll History

Payroll Records Report

Payroll ID	Employee	Pay Period	Gross Pay	Net Pay	HRA Amount	DA Amount	TA Amount	Deductions	Payment Date
5	1	Jul-2026	\$ 39,000.00	\$ 34,500.00	\$ 3,500.00	\$ 1,800.00	\$ 700.00	\$ 0.00	25-Jul-2026
4	1	Jul-2026	\$ 71,000.00	\$ 62,800.00	\$ 6,000.00	\$ 3,000.00	\$ 1,500.00	\$ 0.00	25-Jul-2026
3	1	Jul-2026	\$ 48,000.00	\$ 42,100.00	\$ 4,500.00	\$ 2,200.00	\$ 900.00	\$ 0.00	25-Jul-2026
	2								
	3								
	4								
	5								
2	1	Jul-2026	\$ 62,000.00	\$ 54,200.00	\$ 5,500.00	\$ 2,800.00	\$ 1,200.00	\$ 0.00	25-Jul-2026
	2								
	3								
1	1	Jul-2026	\$ 55,000.00	\$ 48,500.00	\$ 5,000.00	\$ 2,500.00	\$ 1,000.00	\$ 0.00	25-Jul-2026
	2								
	3								

Showing 5 of 5

Show Summary

STEP 6: SELECT ONE FORM

How would you like to create your form?

On my own

From scratch

Import Schema

Using Zia

From a template

Cab Booking

Conf Room Booking

Appointment Booking

Hotel Room Booking

Complaints

50 +
View More

STEP 7: THE SOFTWARE HASE BEEN CREATED.

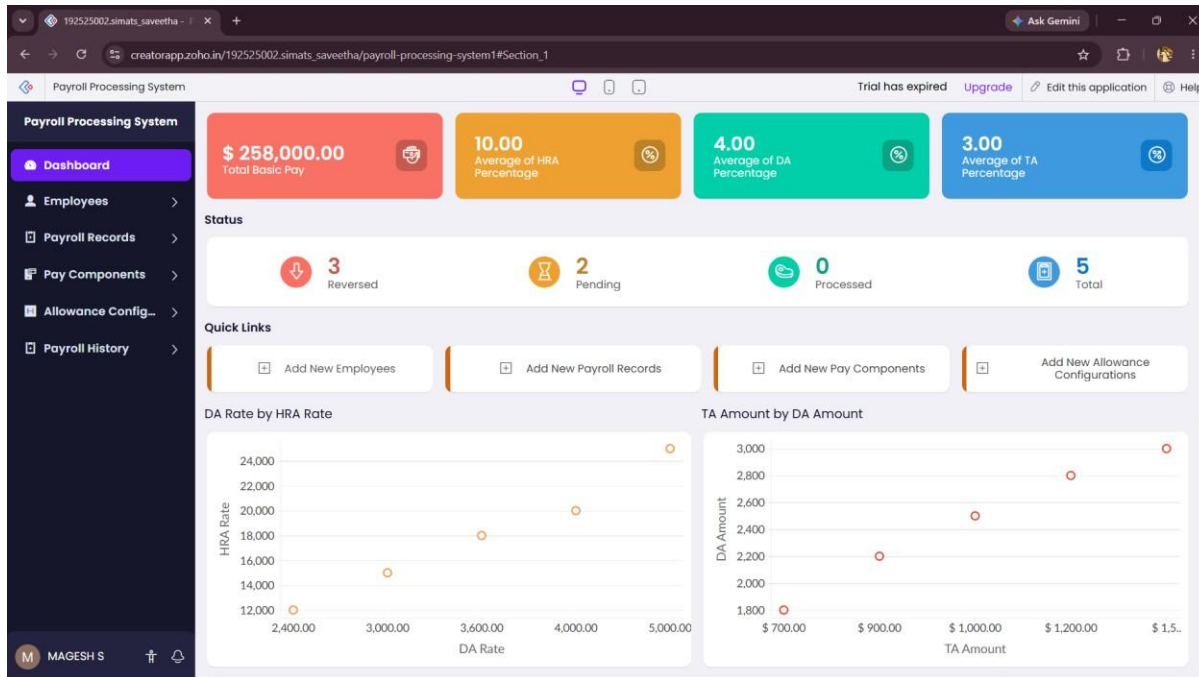
Payroll Processing System

Employees Report

	Employee ID	Employee Name	Experience in Present Org	Overall Experience	Basic Pay	HRA Percentage	DA Percentage	TA Pe
	5	Eva Garcia	4	6	\$ 55,000.00	10.00	4.00	
	4	David Lee	2	3	\$ 48,000.00	10.00	4.00	
	3	Carol Martinez	5	8	\$ 60,000.00	10.00	4.00	
	2	Bob Williams	1	1	\$ 45,000.00	10.00	4.00	
	1	Alice Johnson	3	5	\$ 50,000.00	10.00	4.00	

Showing 5 of 5

Show Summary



Result

Thus, a cloud-based Payroll Processing System was successfully developed and deployed using a Cloud Service Provider. The application accepts employee details, automatically calculates HRA, DA, TA, Gross Pay, and Net Pay, and provides access through the internet, thereby successfully demonstrating the Software as a Service (SaaS) cloud model.

EXP NO 3 :Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

DATE:

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux).

PROCEDURE:

STEP 1:Download VMware workstation and installed as type 2hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

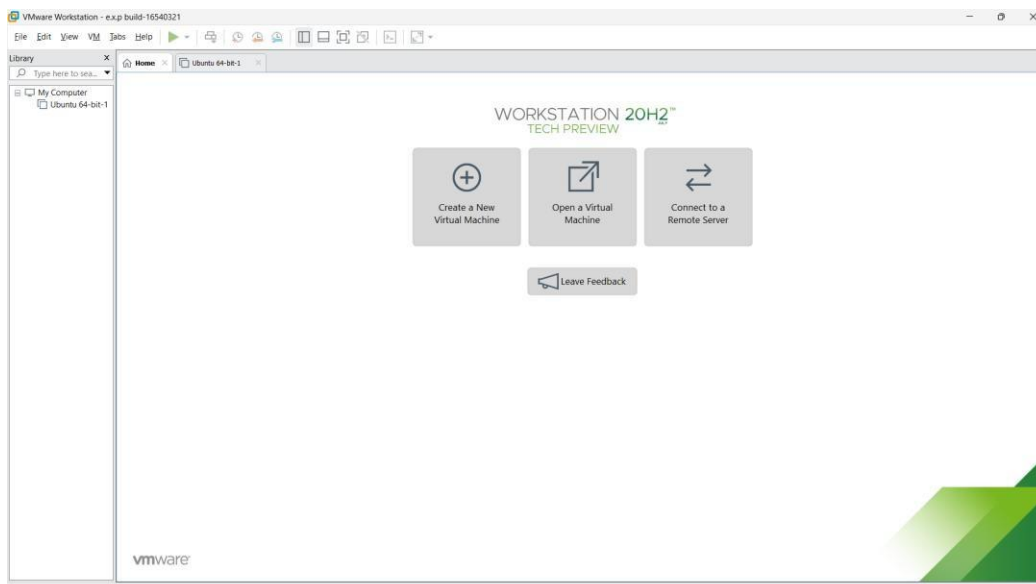
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

IMPLEMENTATION:

STEP 1:DOWLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR

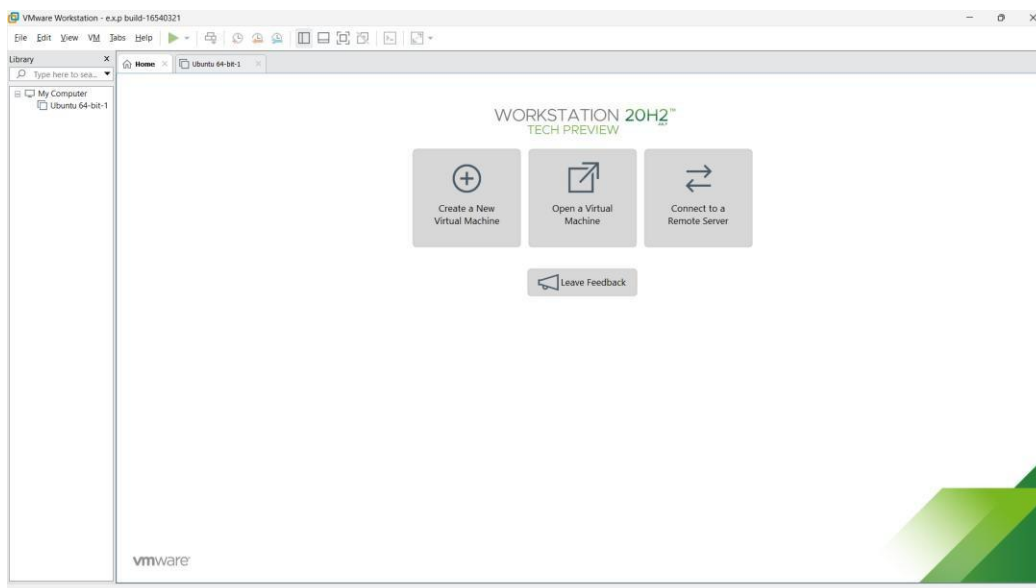


STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

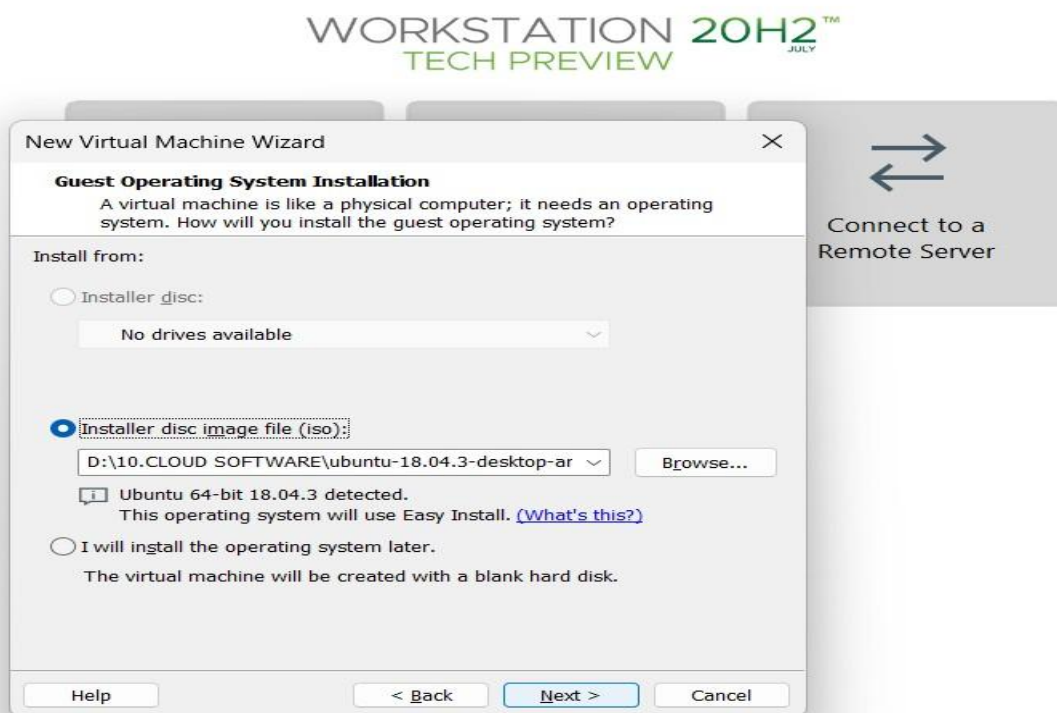
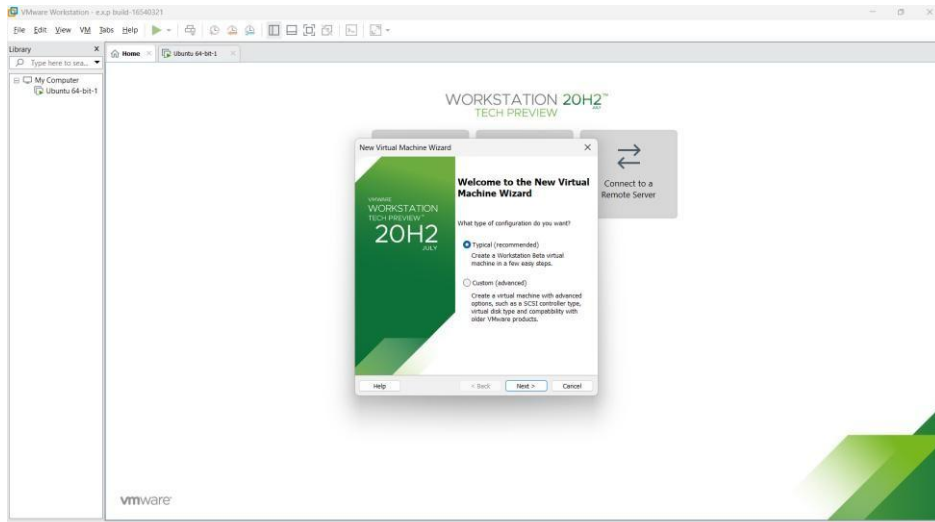
Index of /11.x/x86/release/

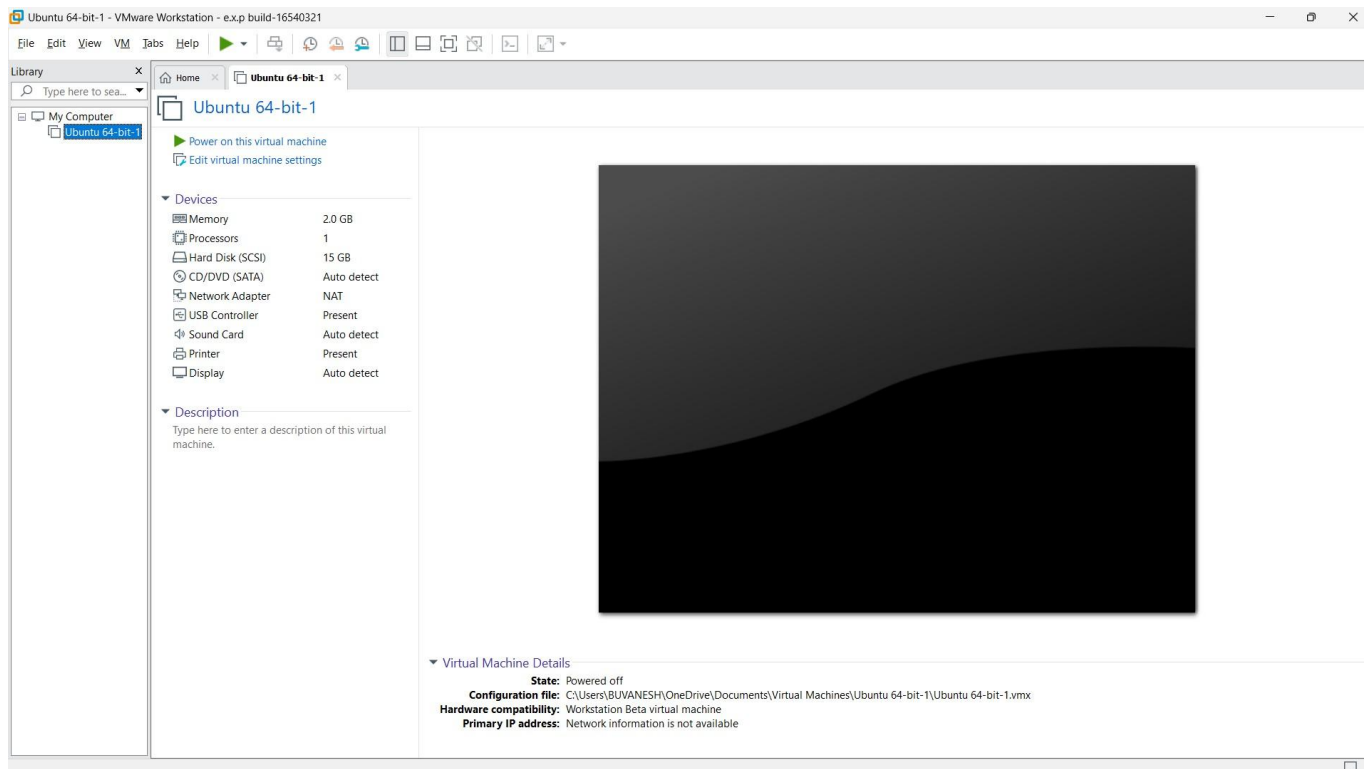
../	09-Feb-2020 11:50	-
distribution_files/	03-Dec-2019 11:14	-
src/	01-Apr-2020 07:49	14757888
Core-11.1.iso	01-Apr-2020 07:49	48
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	50639
Core-11.1.iso.zsync	01-Apr-2020 07:49	14757888
Core-current.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso	01-Apr-2020 07:50	52
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	369358
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	216006656
CorePlus-current.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso	01-Apr-2020 07:50	52
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	68301
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	19922944
TinyCore-current.iso		

STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM



STEP 4: DO THE BASIC CONFIGURATION SETTINGS.





Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux).

Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

AIM:

To create a snapshot of a vm and test it by loading the previous version/cloned vm

PROCEDURE:

STEP 1: GOTO VMWARE WORKSTATION.

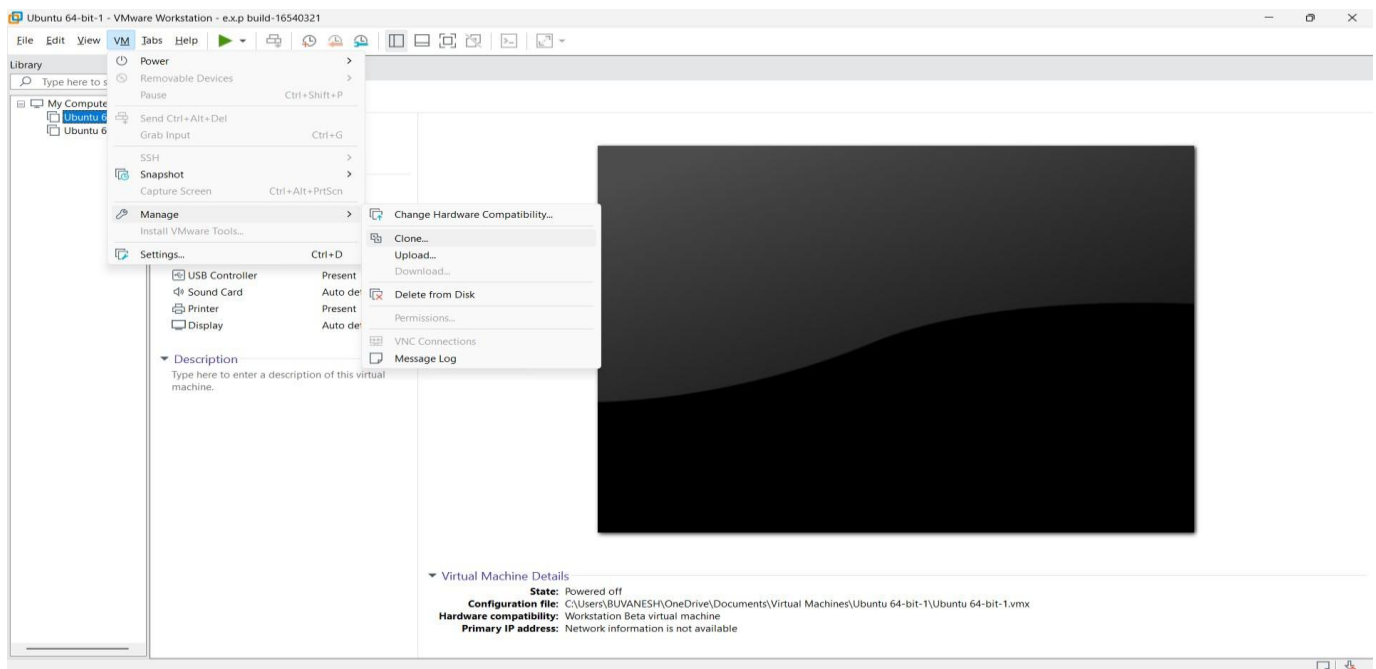
STEP 2: CREATE FILES ON DESKTOP.

STEP 3: CLICK ON VM AND SELECTS SNAPSHOT-> TAKE SNAPSHOT.

STEP 4: SNAPSHOT IS BEING DONE

IMPLEMENTATION:

STEP 1: GO TO VM AND GOTO MANAGE AND CLICK CLONE



STEP 2: CLICK CLONE

